

SECOND SEMESTER M.Tech.(AI)

END-SEMESTER EXAMINATION, April- May 2026

Time- 3hrs

Max. Marks- 40

Course Code: CPAIC03

Course Title- Soft Computing

Note: - Attempt all questions. Missing data/ information if any, may suitably be assumed & mentioned in the answer.

No.	Question	Marks	CO
Q1	Attempt any two parts of the following		
1a	Critically analyze the limitations of Hard Computing and explain how Soft Computing overcomes them with suitable examples	4	I
1b	Differentiate between Fuzzy Set Theory and Probability Theory.	4	I
1c	Discuss recent trends in Soft Computing such as hybrid models and explain their industrial relevance.	4	I
Q2.	Attempt any two parts of the following		
2a	Given: $\mu_A(x) = \begin{cases} x, & 0 \leq x \leq 0.5 \\ 1 - x, & 0.5 < x \leq 1 \end{cases}$ Find: (a) Concentration (b) Dilation Also discuss significance of concentration and dilation with example and figure.	4	II
2b	Analyze the usage of open and closed fuzzy Sets in Fuzzy Logic.	4	II
2c	Consider a fuzzy set A defined as: $A = \{(x_1, 0.3), (x_2, 0.7), (x_3, 1), (x_4, 0.6), (x_5, 0.2)\}$ Find: - Support - Core - Check normality - Crossover point(s) α-cut for $\alpha = 0.7$ (for Memory)	4	II
Q3.	Attempt any two parts of the following		

3a	Discuss local and global connectivity measures of fuzzy graph with their equations.	4	III
3b	Compare Genetic Algorithm and Particle Swarm Optimization.	4	III
3c	Compare Supervised, Unsupervised and Reinforcement Learning along with their applications.	4	III
Q4	Attempt any two parts of the following		
4a	Explain how fuzzy logic can be applied to Word Sense Disambiguation. Discuss the integration of Hindi WordNet with soft computing approaches for semantic analysis.	4	IV
4b	Write the algorithm for WSD using fuzzy graphs and fuzzy graph connectivity measures.	4	IV
4c	Explain fuzzy semantic relations : • Hypernymy / Hyponymy • Meronymy / Holonymy Also discuss their applications in NLP.	4	IV
Q5	Attempt any two parts of the following		
5a	Draw and explain Flow Chart of genetic algorithm iterations.	4	V
5b	Is genetic algorithm admissible? Do they guarantee convergence? Explain why a mutation operator is used in a genetic algorithm. Discuss two methods to select chromosome from sampling space.	4	V
5c	Discuss Particle Swarm Optimization method. Solve following problem using PSO. • Objective Function: $Maximize f(x) = 1 + 2x - x^2$ • Let our Control parameters be • $W = 0.70$; $C_1 = 0.20$; $C_2 = 0.60$; $n = 5$ (Five Swarm Particles) $E = 4$ iterations (for simplicity) • Random numbers used for updating Velocity of particles be: $r_1 = [0.4657, 0.8956, 0.3877, 0.4902, 0.5039]$ (each one corresponding each particle) $r_2 = [0.5319, 0.8185, 0.8331, 0.7677, 0.1708]$	4	V